

11	C	Option C (correct): $g = -\frac{d\phi}{dr} \Rightarrow d\phi = -\int g \, dr$ Hence, the area under the g-r graph gives the change in the gravitational potential ϕ. Options A and B (incorrect): The total gravitational potential between the two planets is always negative. Gravitational potential is zero only at infinity. Option D (incorrect): The gradient of the graph does not give any meaningful quantity.
12	A	The gravitational force on each star provides the centripetal force for the star to orbit